

## References

### Articles

1. Illa, B., **Malkoti, A.**, Hanasoge, S., Plessix, R.-E., Chandran, A., (2025). "Homogenization of Elastic Wave Equation using Renormalization Group Theory". In: *EarthArxiv (Preprint)*. DOI: 10.31223/X5JJ2N.
2. Kumawat, S., **Malkoti, A.**, Vishwakarma, S. K., (2024). "A cell-centered implicit finite difference scheme to study wave propagation in acoustic media: A numerical modeling". In: *Journal of Sound and Vibration* 590. IF=4.9, p. 118601. ISSN: 0022460X. DOI: 10.1016/j.jsv.2024.118601.
3. **Malkoti, A.**, Hanasoge, S. M., Plessix, R.-É., (2022). "Upscaling acoustic wave equation using renormalization group theory". In: *Geophysics* 87.4. IF=3.264, T281–T290. ISSN: 0016-8033, 1942-2156. DOI: 10.1190/geo2021-0572.1.
4. Vadapalli, U., **Malkoti, A.**, Vedanti, N., Dimri, V. P., Guruhappa, H., (2022). "Field-scale and site-specific permeability of sandstone reservoirs using a stabilized Monte Carlo algorithm: A case study". In: *Geophysics* 87.6. IF=3.2, B337–B350. ISSN: 0016-8033, 1942-2156. DOI: 10.1190/geo2022-0255.1.
5. Li, L., Tan, J., Zhang, D., **Malkoti, A.**, Abakumov, I., Xie, Y., (2021). "FDwave3D: a MATLAB solver for the 3D anisotropic wave equation using the finite-difference method". In: *Computational Geosciences* 25.5. IF=2.948, pp. 1565–1578. ISSN: 1420-0597, 1573-1499. DOI: 10.1007/s10596-021-10060-3.
6. **Malkoti, A.**, Datta, A., Hanasoge, S. M., (2021). "Rayleigh-wave H/V ratio measurement from ambient noise cross-correlations and its sensitivity to V P: a numerical study". In: *Geophysical Journal International* 227.1. IF=3.352, pp. 472–482. ISSN: 0956-540X, 1365-246X. DOI: 10.1093/gji/ggab228.
7. **Malkoti, A.**, Vedanti, N., Tiwari, R. K., (2019). "A highly efficient implicit finite difference scheme for acoustic wave propagation". In: *Journal of Applied Geophysics* 161. IF=1.845, pp. 204–215. ISSN: 09269851. DOI: 10.1016/j.jappgeo.2018.12.017.
8. Vedanti, N., **Malkoti, A.**, (2019b). "Viscoelastic seismic modeling and Q estimation for an attenuating media." In: *Journal of Indian Geophysical Union* 23.1, pp. 3–9.
9. **Malkoti, A.**, Vedanti, N., Tiwari, R. K., (2018). "An algorithm for fast elastic wave simulation using a vectorized finite difference operator". In: *Computers & Geosciences* 116. IF=5.168, pp. 23–31. ISSN: 00983004. DOI: 10.1016/j.cageo.2018.04.002.
10. Vedanti, N., **Malkoti, A.**, Pandey, O. P., Shrivastava, J. P., (2018). "Ultrasonic P- and S-Wave Attenuation and Petrophysical Properties of Deccan Flood Basalts, India, as Revealed by Borehole Studies". In: *Pure and Applied Geophysics* 175.8. IF=2.641, pp. 2905–2930. ISSN: 0033-4553, 1420-9136. DOI: 10.1007/s00024-018-1817-x.
11. **Malkoti, A.**, Vedanti, N., Bhattacharaya, A. K., (2017). "Inversion of well log data using improved shale model for determination of petrophysical parameters." In: *Journal of Indian Geophysical Union* 21.2.

### Conferences

1. Kalyankar, S., Lalngaihawma, J., **Malkoti, A.**, Kumar, P., Vedanti, N., (2025). "Seismic Fault Detection on Raniganj Data using Deep Learning". In: Beneath the Surface: Innovations in Geoscience. Hyderabad, India (18-20 June): Society of Exploration Geophysicist.
2. Kumar, P., Kumar, K., George, N., Ahmed, S., **Malkoti, A.**, Vedanti, N., (2025). "Mapping a sedimentary sub-basin in the eastern Raniganj Basin, India using geophysical data: A potential site for CO<sub>2</sub> sequestration". In: Beneath the Surface: Innovations in Geoscience. Hyderabad, India (18-20 June): Society of Exploration Geophysicist.
3. Lalngaihawma, J., Kalyankar, S., **Malkoti, A.**, Vedanti, N., (2025). "Seismic fault identification using a hybrid ResNet-UNet architecture". In: International Meeting for Applied Geoscience & Energy (IMAGE). Houston, Texas, USA (25-28 Aug): Society of Exploration Geophysicists and American Association of Petroleum Geologists.
4. Zia, S., **Malkoti, A.**, Vedanti, N., (2025a). "Efficient monitoring of a geological carbon sequestration process through accelerated poroviscoelastic wave modeling with Convolutional-PML boundary conditions". In: International Meeting for Applied Geoscience & Energy (IMAGE), Houston, Texas, USA (25-28 Aug): Society of Exploration Geophysicist.
5. Zia, S., **Malkoti, A.**, Vedanti, N., (2025b). "Poro-viscoelastic simulation and its application for seismic CO<sub>2</sub> monitoring in the Gandhar oilfield, India". In: Beneath the Surface: Innovations in Geoscience. Hyderabad, India (18-20 June): Society of Exploration Geophysicist.

6. Kumar, P., George, N., Ahmed, S., **Malkoti, A.**, Sinha, S., Vedanti, N., Dhar, R., (2024). "Structural model of coal-bearing Raniganj Formation derived from petrophysical data in the Raniganj (E) block, India". In: National Conference on Rock Deformation and Structures (RDS). IISER Bhopal (22-24 October).
7. Kumar, P., **Malkoti, A.**, Vedanti, N., (2024). "Geological sequestration of the CO<sub>2</sub> in the potential coal seams from the Raniganj basin". In: Role of Geosciences in Carbon Storage. Mumbai, India (19-21 March): Society of Exploration Geophysicist.
8. Singh, S., Kalyankar, S., **Malkoti, A.**, Vedanti, N., (2024). "Application of fractional Gaussian fractal-based stochastic seismic inversion to F3-Netherland dataset". In: Fourth International Meeting for Applied Geoscience & Energy. Houston, Texas (26-29 Aug): Society of Exploration Geophysicists and American Association of Petroleum Geologists, pp. 109–113. DOI: 10.1190/image2024-4100573.1.
9. Vadapalli, U., **Malkoti, A.**, (2024). "Challenges in permeability prediction concerning geological carbon sequestration: A case study from Ankleshwar oil field, India". In: Role of Geoscience in Carbon Storage. Mumbai, India (19-21 March): Society of Exploation Geophysicist.
10. **Malkoti, A.**, Vedanti, N., (2023). "Application of Seismic Exploration Techniques in Exploration of Unconventional Hydrocarbons and CCS projects". In: International Conference on Petroleum, Hydrogen and Decarbonization. IIT Guwahati, Assam, India (3-5 Nov).
11. **Malkoti, A.**, Hanasoge, S. M., Edouard Plessix, R., (2022). "Upscaling acoustic wave equation using renormalization group theory". In: *Second International Meeting for Applied Geoscience & Energy*. Second International Meeting for Applied Geoscience & Energy. Houston, Texas (28 Aug - 1 Sept): Society of Exploration Geophysicists and American Association of Petroleum Geologists, pp. 2507–2511. DOI: 10.1190/image2022-3744890.1.
12. **Malkoti, A.**, Datta, A., Hanasoge, S., (2020). "H/V amplitude ratio measurement using multicomponent ambient noise cross-correlations and its relationship to V<sub>p</sub>/V<sub>s</sub>." In: *AGU Fall Meeting 2020*. AGU Fall Meeting 2020. Online.
13. **Malkoti, A.**, Vedanti, N., (2019). "Efficient seismic simulation for a highly attenuating media," in: *International Union of Geodesy and Geophysics 27th General assembly*. International Union of Geodesy and Geophysics 27th General assembly. Montreal, Canada.
14. Vedanti, N., **Malkoti, A.**, (2019a). "Petrophysical characterization of two formations of Deccan flood basalt for improved sub-basalt imaging". In: *International Union of Geodesy and Geophysics 27th General assembly*. International Union of Geodesy and Geophysics 27th General assembly. Montreal, Canada.
15. Malik, R., **Malkoti, A.**, Vedanti, N., Dimri, V., (2017). "1-D Full waveform inversion using micro genetic algorithm". In: Association of Exploration Geophysicist 39th Annual Convention. Banaras Hindu University, Varanasi, India.
16. **Malkoti, A.**, Vedanti, N., Tiwari, R., (2017). "High fidelity numerical simulation for scaler wave". In: *Association of Exploration Geophysicist 39th Annual Convention*. Association of Exploration Geophysicist 39th Annual Convention. Banaras Hindu University, Varanasi, India.
17. **Malkoti, A.**, Vedanti, N., Kunagu, P., Tiwari, R., (2015). "Modeling viscoelastic seismic wave propagation in Deccan flood basalt, western India". In: *SEG Technical Program Expanded Abstracts 2015*. SEG Technical Program Expanded Abstracts 2015. New Orleans, Louisiana (19-21 Aug): Society of Exploration Geophysicists, pp. 3764–3768. DOI: 10.1190/segam2015-5898555.1.
18. Vedanti, N., Lakshmi, K., Dutta, S., **Malkoti, A.**, Pandey, O., (2015). "Investigation of Petrophysical Properties and Ultrasonic P-and S- Wave attenuation in Deccan Flood Basalts, India". In: *SEG Technical Program Expanded Abstracts 2015*. SEG Technical Program Expanded Abstracts 2015. New Orleans, Louisiana (19-21 Aug): Society of Exploration Geophysicists, pp. 3274–3278. DOI: 10.1190/segam2015-5858683.1.
19. Vedanti, N., **Malkoti, A.**, Datta, S., Pandey, O., (2015). "Elastic properties of ambenali and poladpur formations of DVP: New findings," in: *National conference on Paleogene of the Indian Subcontinent*, National conference on Paleogene of the Indian Subcontinent, Birbal Sahni Institute for Palaeobotany, Lucknow (23- 24 April).
20. **Malkoti, A.** (2014). "Inversion of well log data for better determination of petrophysical parameters". In: *Indian Geophysical Union 51st Annual convention*. Indian Geophysical Union 51st Annual convention. Kurukshetra University, Haryana, India.